

German Accident Data Platform

API Documentation

Version 2.0.0

Base URL: <http://127.0.0.1:8000>

Interactive Docs: <http://127.0.0.1:8000/docs> (Swagger UI)

OpenAPI Schema: <http://127.0.0.1:8000/openapi.json>

All endpoints use HTTP GET and return JSON. Every data response includes a provenance block with source attribution and licence (dl-de/by-2-0).

Endpoint Overview

Endpoint	Category	Description
GET /	Meta	Service index with version and endpoint list
GET /health	Meta	Liveness check, returns {status: ok}
GET /aggregates/accidents	Count	Total accident count for any filter combination
GET /aggregates/accidents/by-region	Ranking	Counts grouped by state/district/municipality
GET /aggregates/hotspots	Spatial	Severity-ranked crash clusters on a grid
GET /accidents	List	Paginated individual accident rows
GET /accidents/near	Spatial	Accidents within a radius of a lat/lon point
GET /regions/choropleth	GeoJSON	District polygons with values for choropleth maps
GET /stats/first-year	Temporal	Earliest data year overall or per state
GET /stats/trend	Trend	Accident counts per year as a time series
GET /metadata/sources	Provenance	Data sources and their licences
GET /import-runs	Provenance	Every import run with timestamps and row counts

Endpoint Reference

GET /aggregates/accidents

Returns the total accident count for the given filter combination. This is the primary count endpoint.

Parameters:

Parameter	Type	Required	Description
state	string	No	State code (SN, BE, NW) or 2-digit AGS (14)
region	string	No	AGS prefix (2/5/8 digits); ignored if state given
year	integer	No	Filter by year (2016–2024)
month	integer	No	Filter by month (1–12)
category	integer	No	1=fatal, 2=serious injury, 3=light injury
pedestrian	boolean	No	Only crashes involving pedestrians
bicycle	boolean	No	Only crashes involving cyclists
car	boolean	No	Only crashes involving cars
motorcycle	boolean	No	Only crashes involving motorcycles
goods	boolean	No	Only crashes involving goods vehicles

Example request:

```
GET /aggregates/accidents?state=SN&year=2023&category=1
```

Example response:

```
{ "query": { "state": "SN", "year": 2023, "category": 1 },  
  "accident_count": 162,  
  "provenance": [{ "source": "Unfallatlas", "license": "dl-de/by-2-0", ... }] }
```

GET /aggregates/accidents/by-region

Returns accident counts grouped by region level, sorted by count. Used for ranking tables.

Parameters:

Parameter	Type	Required	Description
level	string	No	Grouping level: state, district, or municipality (default: state)
year	integer	No	Filter by year
category	integer	No	1=fatal, 2=serious, 3=light
state	string	No	Limit to a state (e.g. SN, BE, NW)
region	string	No	Limit to an AGS prefix
order	string	No	desc (default) = highest first, asc = lowest first
limit	integer	No	Max results (default: 20, max: 1000)
pedestrian	boolean	No	Only pedestrian-involved crashes
bicycle	boolean	No	Only cyclist-involved crashes

Example request:

```
GET /aggregates/accidents/by-region?level=district&year=2023&state=SN&limit=3
```

Example response:

```
[{ "ags": "14612", "name": "Dresden", "level": "district",  
  "year": 2023, "accident_count": 341 }, ...]
```

GET /aggregates/hotspots

Identifies accident hotspots by clustering crashes on a spatial grid and ranking them by a severity-weighted score (fatal=10, serious=3, light=1).

Parameters:

Parameter	Type	Required	Description
precision	integer	No	Grid resolution: 2=~1.1km, 3=~110m (default), 4=~11m
min_count	integer	No	Minimum crashes in a cell to qualify (default: 3, min: 1)
year	integer	No	Filter by year
state	string	No	Limit to a state
region	string	No	Limit to an AGS prefix
category	integer	No	1=fatal, 2=serious, 3=light
weighted	boolean	No	Rank by severity score (true) or raw count (false)
limit	integer	No	Max hotspots returned (default: 20, max: 500)

Example request:

```
GET /aggregates/hotspots?state=BY&year=2023&category=1&precision=2&min_count=2
```

GET /accidents

Returns individual accident rows, paginated. Each row includes coordinates, severity, participant flags, and the 8-digit municipality AGS.

Parameters:

Parameter	Type	Required	Description
state	string	No	State code or 2-digit AGS
region	string	No	AGS prefix (2/5/8 digits)
year	integer	No	Filter by year
month	integer	No	Filter by month (1–12)
weekday	integer	No	Filter by weekday (1–7)
hour	integer	No	Filter by hour (0–23)
category	integer	No	1=fatal, 2=serious, 3=light
limit	integer	No	Page size (default: 50, max: 1000)
offset	integer	No	Pagination offset (default: 0)

Example request:

```
GET /accidents?state=BE&year=2023&category=1&limit=5
```

GET /accidents/near

Spatial query: returns all accidents within a given radius of a point, sorted by distance.

Parameters:

Parameter	Type	Required	Description
lat	float	Yes	Latitude (47.0–55.5, WGS84)
lon	float	Yes	Longitude (5.5–15.5, WGS84)
radius_km	float	No	Search radius in km (default: 2.0, max: 50)
year	integer	No	Filter by year
category	integer	No	1=fatal, 2=serious, 3=light
limit	integer	No	Max results (default: 500)

Example request:

```
GET /accidents/near?lat=52.52&lon=13.405&radius_km=1&year=2023
```

GET /regions/choropleth

Returns a GeoJSON FeatureCollection of district polygons with accident counts or rate values attached, ready for rendering as a choropleth map.

Parameters:

Parameter	Type	Required	Description
metric	string	No	count (default) or rate10k
year	integer	Yes	Year for accident counts or indicator
state	string	No	Limit to a state (returns only that state's districts)
category	integer	No	1=fatal, 2=serious, 3=light

Example request:

GET /regions/choropleth?metric=count&year=2023&state=SN&category=2

GET /stats/trend

Returns a time series of accident counts per year, optionally filtered by state and severity.

Parameters:

Parameter	Type	Required	Description
state	string	No	Limit to a state
category	integer	No	1=fatal, 2=serious, 3=light

Example request:

```
GET /stats/trend?state=SN&category=1
```

Example response:

```
{ "state": "SN", "series": [
  { "year": 2016, "accident_count": 185 },
  { "year": 2017, "accident_count": 171 }, ... ],
  "provenance": [...] }
```

GET /stats/first-year

Returns the earliest year for which accident data is available.

Parameter	Type	Required	Description
state	string	No	Limit to a state

```
GET /stats/first-year?state=SN → { "state": "SN", "first_year": 2016 }
```

GET /metadata/sources

Returns the list of integrated data sources with their licences.

```
GET /metadata/sources
```

```
[{ "source": "Unfallatlas", "license": "dl-de/by-2-0",
  "license_url": "https://www.govdata.de/dl-de/by-2-0" }, ...]
```

GET /import-runs

Returns the full import audit trail: every ETL run with source, timestamp, file name, and row count.

Error Response Format

All errors return a consistent JSON structure:

```
{ "error": true, "status": 400,  
  "detail": "Unknown state 'XX'.",  
  "path": "/aggregates/accidents" }
```

Common status codes: 400 (invalid parameter), 404 (resource not found), 422 (validation error), 500 (server error).

Severity Categories

Value	Label	Description
1	Fatal	Accident with at least one fatality
2	Serious	Serious injury, no fatality
3	Light	Light injury only

State Codes

The API accepts both 2-letter abbreviations and 2-digit AGS codes:

Code	AGS	State	Code	AGS	State
SH	01	Schleswig-Holstein	RP	07	Rheinland-Pfalz
HH	02	Hamburg	BW	08	Baden-Württemberg
NI	03	Niedersachsen	BY	09	Bayern
HB	04	Bremen	SL	10	Saarland
NW	05	Nordrhein-Westfalen	BE	11	Berlin
HE	06	Hessen	BB	12	Brandenburg
MV	13	Mecklenburg-Vorpommern	SN	14	Sachsen
ST	15	Sachsen-Anhalt	TH	16	Thüringen